

Session 14: Debugging & Best Practices Topics: Common syntax/runtime errors. Importance of commenting & indentation.

Tasks

- 1) Rewrite the following code to improve its indentation and add comments explaining each step, so that a beginner can understand what it does:
`function isEven(num){if(num%2==0){return true;}else{return false;}}`

Code-

```
#include <stdio.h>
```

```
int isEven(int num)
{
```

```
    if (num % 2 == 0)
    {
        return 1; // 1 represents "true" in C (even number)
    }
    else
    {
        return 0; // 0 represents "false" in C (odd number)
    }
}
```

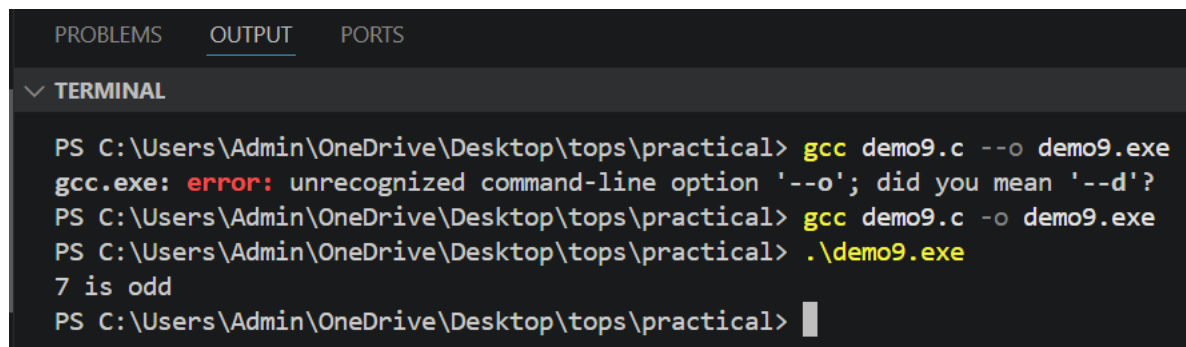
```
int main()
{
    int number = 7;

    int result = isEven(number);
```

```
    if (result)
    {
        printf("%d is even\n", number);
    }
    else
    {
        printf("%d is odd\n", number);
    }

    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
▼ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc demo9.c --o demo9.exe
gcc.exe: error: unrecognized command-line option '--o'; did you mean '--d'?
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc demo9.c -o demo9.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\demo9.exe
7 is odd
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```

2) Write a function format Followers Count(count) that takes a number and returns a formatted string like Instagram: 1500 as '1.5K', 1200000 as '1.2M', and numbers below 1000 as-is. Add clear comments and use proper indentation.

Code-

```
#include <stdio.h>
```

```
void formatFollowersCount(long count, char *result)
```

```
{
```

```
    if (count >= 1000000)
```

```
    {
```

```
        double millions = count / 1000000.0;
```

```
        sprintf(result, "%.1fM", millions);
```

```
    }
```

```
    else if (count >= 1000)
```

```
    {
```

```
        double thousands = count / 1000.0;
```

```
        sprintf(result, "%.1fK", thousands);
    }

    else
    {
        sprintf(result, "%ld", count);
    }
}

int main()
{

    char formatted[20];

    long testValues[] = {950, 1500, 1200000, 999, 1000000, 25000};
    int numTests = sizeof(testValues) / sizeof(testValues[0]);

    for (int i = 0; i < numTests; i++)
    {

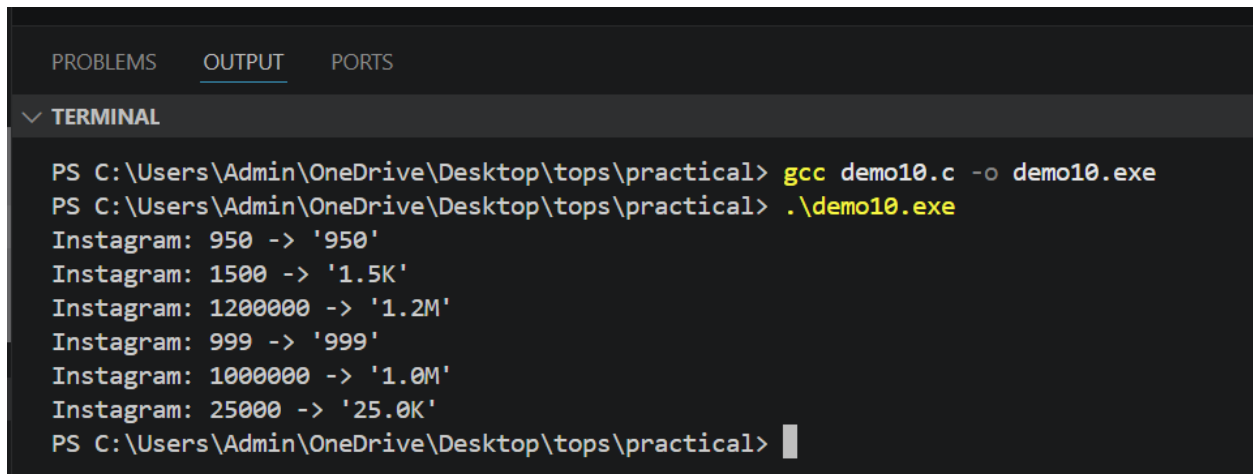
        formatFollowersCount(testValues[i], formatted);

        printf("Instagram: %ld -> '%s'\n", testValues[i], formatted);
    }

    return 0;
```

}

Output:



```
PROBLEMS  OUTPUT  PORTS
▼ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc demo10.c -o demo10.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\demo10.exe
Instagram: 950 -> '950'
Instagram: 1500 -> '1.5K'
Instagram: 1200000 -> '1.2M'
Instagram: 999 -> '999'
Instagram: 1000000 -> '1.0M'
Instagram: 25000 -> '25.0K'
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```

3) You are given a piece of JavaScript code that should print all even numbers from 1 to 10, but it doesn't work as expected:

for (let i = 1; i <= 10; i++) {
 if (i % 2 = 0) {
 console.log(i)
 }
}

Find and fix the error, then rewrite the code with good indentation and at least one comment explaining the logic.

Code-

- `int i = 1` — `let` is JavaScript syntax, not C. In C you just declare the type: `int i = 1`.
- `i % 2 = 0` — `=` is assignment, not comparison. Needs `==`.
- `printf(i)` — `printf` needs a format string as the first argument (like `"%d\n"`) to know how to display the value; passing just `i` won't correctly print an integer.
- The loop also needs to live inside a `main()` function to compile as a valid C program.

```
#include <stdio.h>
```

```
int main()
{
    for (int i = 1; i <= 10; i++)
    {
        if (i % 2 == 0)
        {
            printf("%d\n", i);
        }
    }

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc demo11.c -o demo11.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\demo11.exe
2
4
6
8
10
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```